

## Modal Logic: Tableaux Examples

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Example 13:  $\neg\Box(A \wedge \neg\Box A)$  in K

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1.  $\mathbb{F} 1, \neg \Box(A \wedge \neg \Box A)$  ✓ Assumption
2.  $\mathbb{T} 1, \Box(A \wedge \neg \Box A) \quad \neg \mathbb{F} 1$

Assume the formula is false at world 1. False  $\neg$  gives us  $\Box(A \wedge \neg \Box A)$  true at world 1.

### Example 13: $\neg\Box(A \wedge \neg\Box A)$ in K

Line 2 is True  $\Box$ , which is a reusable rule — it can only fire when we have an accessible world.

But no accessible worlds have been introduced, and no other rules apply. The tree stays open.

The countermodel: a single world  $w_1$  with no accessible worlds at all.

$\Box(A \wedge \neg\Box A)$  is vacuously true there (no accessible worlds to check), so  $\neg\Box(A \wedge \neg\Box A)$  is false at  $w_1$ , as required.

Example 14:  $\neg \Box(A \wedge \neg \Box A)$  in KT

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## Example 14: $\neg\Box(A \wedge \neg\Box A)$ in KT

In **KT** every world accesses itself, so the True  $\Box$  rule now has somewhere to fire. Let's start from where we were.

1.  $\models 1, \neg \Box(A \wedge \neg \Box A)$  ✓ Assumption
2.  $\Vdash 1, \Box(A \wedge \neg \Box A) \rightarrow \models 1$

In KT, world 1 accesses itself, so we can apply the True  $\Box$  rule at world 1.



1.  $\mathbb{F} 1, \neg \Box(A \wedge \neg \Box A)$  ✓ Assumption
2.  $\mathbb{T} 1, \Box(A \wedge \neg \Box A)$   $\neg \mathbb{F} 1$
3.  $\mathbb{T} 1, A \wedge \neg \Box A$   $\mathbb{T} \Box 2$

The  $\mathbb{T} \Box$  rule (line 2) gives us  $A \wedge \neg \Box A$  true at world 1: since  $\Box(A \wedge \neg \Box A)$  is true at 1 and 1 accesses itself, the formula inside the box must hold at 1.

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|----|---------------------------------------------------|-----------------------|
| 1. | $\mathbb{F} 1, \neg \Box(A \wedge \neg \Box A)$ ✓ | Assumption            |
| 2. | $\mathbb{T} 1, \Box(A \wedge \neg \Box A)$        | $\neg \mathbb{F} 1$   |
| 3. | $\mathbb{T} 1, A \wedge \neg \Box A$ ✓            | $\mathbb{T} \Box 2$   |
| 4. | $\mathbb{T} 1, A$                                 | $\wedge \mathbb{T} 3$ |
| 5. | $\mathbb{T} 1, \neg \Box A$                       | $\wedge \mathbb{T} 3$ |

True  $\wedge$  (line 3) gives both conjuncts:  $A$  true at 1 (line 4) and  $\neg \Box A$  true at 1 (line 5). Line 3 is checked.

|    |                                                   |                       |
|----|---------------------------------------------------|-----------------------|
| 1. | $\mathbb{F} 1, \neg \Box(A \wedge \neg \Box A)$ ✓ | Assumption            |
| 2. | $\mathbb{T} 1, \Box(A \wedge \neg \Box A)$        | $\neg \mathbb{F} 1$   |
| 3. | $\mathbb{T} 1, A \wedge \neg \Box A$ ✓            | $\mathbb{T} \Box 2$   |
| 4. | $\mathbb{T} 1, A$                                 | $\wedge \mathbb{T} 3$ |
| 5. | $\mathbb{T} 1, \neg \Box A$ ✓                     | $\wedge \mathbb{T} 3$ |
| 6. | $\mathbb{F} 1, \Box A$                            | $\neg \mathbb{T} 5$   |
| 7. | $\mathbb{F} 1.1, A$                               | $\Box \mathbb{F} 6$   |

True  $\neg$  (line 5) gives  $\Box A$  false at 1 (line 6). False  $\Box$  (line 6) introduces world 1.1 with  $A$  false there (line 7). Lines 5 and 6 are checked.

|    |                                                   |                       |
|----|---------------------------------------------------|-----------------------|
| 1. | $\mathbb{F} 1, \neg \Box(A \wedge \neg \Box A)$ ✓ | Assumption            |
| 2. | $\mathbb{T} 1, \Box(A \wedge \neg \Box A)$        | $\neg \mathbb{F} 1$   |
| 3. | $\mathbb{T} 1, A \wedge \neg \Box A$ ✓            | $\mathbb{T} \Box 2$   |
| 4. | $\mathbb{T} 1, A$                                 | $\wedge \mathbb{T} 3$ |
| 5. | $\mathbb{T} 1, \neg \Box A$ ✓                     | $\wedge \mathbb{T} 3$ |
| 6. | $\mathbb{F} 1, \Box A$                            | $\neg \mathbb{T} 5$   |
| 7. | $\mathbb{F} 1.1, A$                               | $\Box \mathbb{F} 6$   |
| 8. | $\mathbb{T} 1.1, A \wedge \neg \Box A$ ✓          | $\Box \mathbb{T} 2$   |
| 9. | $\mathbb{T} 1.1, A$                               | $\wedge \mathbb{T} 8$ |



True  $\Box$  (line 2) is reusable: now that world 1.1 exists, it fires there too, giving  $A \wedge \neg \Box A$  true at 1.1 (line 8). True  $\wedge$  on line 8 gives  $A$  true at 1.1 (line 9) — but line 7 says  $A$  is false at 1.1. Contradiction. The branch closes. This formula is **valid** in **KT**.

## Why is this valid?

- Assume  $\Box(A \wedge \neg\Box A)$  is true at  $w_1$ : at every accessible world,  $A$  is true and  $\Box A$  is false.
- By the T condition,  $w_1$  accesses itself, so  $A \wedge \neg\Box A$  holds at  $w_1$  itself.
- In particular,  $\neg\Box A$  is true at  $w_1$ : so  $\Box A$  is false at  $w_1$ .
- But  $\Box A$  false at  $w_1$  means there is some accessible world where  $A$  fails.
- That contradicts our assumption that  $A$  holds at every accessible world (including  $w_1$  itself, by the T condition).

The T condition is essential: it forces  $w_1$  to be a witness against its own assumption. In K, a world with no accessible worlds escapes this argument entirely.